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Part 1 - Initial setup

1. Configure the appropriate hostname on each router/switch.
2. Configure the enable secret `jeremysitlab` on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
3. Configure the user account `cisco` with secret `ccna` on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
4. Configure the console line to require login with a local user account. Set a 30-minute inactivity timeout. Enable synchronous logging.

Part 2 – VLANs, Layer-2 EtherChannel

1. In Office A, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-A1 and DSW-A2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel.
2. In Office B, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-B1 and DSW-B2 using an open standard protocol. Both switches should actively try to form an EtherChannel.
3. Configure all links between Access and Distribution switches, including the EtherChannels, as trunk links.
 - a. Explicitly disable DTP on all ports.
 - b. Set each trunk's native VLAN to VLAN 1000 (unused).
 - c. In Office A, allow VLANs 10, 20, 40, and 99 on all trunks.
 - d. In Office B, allow VLANs 10, 20, 30, and 99 on all trunks.
4. Configure one of each office's Distribution switches as a VTPv2 server. Use domain name **JeremysITLab**.
 - a. Verify that other switches join the domain.
 - b. Configure all Access switches as VTP clients.

5. In Office A, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.

- a.** VLAN 10: PCs
- b.** VLAN 20: Phones
- c.** VLAN 40: Wi-Fi
- d.** VLAN 99: Management

6. In Office B, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.

- a.** VLAN 10: PCs
- b.** VLAN 20: Phones
- c.** VLAN 30: Servers

d. VLAN 99: Management

7. Configure each Access switch's access port.

- a.** LWAPs will not use FlexConnect
- b.** PCs in VLAN 10, Phones in VLAN 20
- c.** SRV1 in VLAN 30
- d.** Manually configure access mode and explicitly disable DTP

8. Configure ASW-A1's connection to WLC1:

- a.** It must support the Wi-Fi and Management VLANs.
- b.** The Management VLAN should be untagged.
- c.** Disable DTP.

9. Administratively disable all unused ports on Access and Distribution switches.

Part 3 – IP Addresses, Layer-3 EtherChannel, HSRP

1. Configure the following IP addresses on R1's interfaces and enable them:

- a. G0/0/0: DHCP client
- b. G0/1/0: DHCP client
- c. G0/0: 10.0.0.33/30
- d. G0/1: 10.0.0.37/30
- e. Loopback0: 10.0.0.76/32

2. Enable IPv4 routing on all Core and Distribution switches.

3. Create a Layer-3 EtherChannel between CSW1 and CSW2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel. Configure the following IP addresses:

- a. CSW1 PortChannel1: 10.0.0.41/30
- b. CSW2 PortChannel1: 10.0.0.42/30

4. Configure the following IP addresses on CSW1. Disable all unused interfaces.

- a. G1/0/1: 10.0.0.34/30
- b. G1/1/1: 10.0.0.45/30
- c. G1/1/2: 10.0.0.49/30
- d. G1/1/3: 10.0.0.53/30
- e. G1/1/4: 10.0.0.57/30
- f. Loopback0: 10.0.0.77/32

5. Configure the following IP addresses on CSW2. Disable all unused interfaces.

- a. G1/0/1: 10.0.0.38/30
- b. G1/1/1: 10.0.0.61/30
- c. G1/1/2: 10.0.0.65/30
- d. G1/1/3: 10.0.0.69/30
- e. G1/1/4: 10.0.0.73/30
- f. Loopback0: 10.0.0.78/32

6. Configure the following IP addresses on DSW-A1:

- a. G1/1/1: 10.0.0.46/30
- b. G1/1/2: 10.0.0.62/30
- c. Loopback0: 10.0.0.79/32

7. Configure the following IP addresses on DSW-A2:

- a. G1/1/1: 10.0.0.50/30
- b. G1/1/2: 10.0.0.66/30
- c. Loopback0: 10.0.0.80/32

8. Configure the following IP addresses on DSW-B1:

- a. G1/1/1: 10.0.0.54/30
- b. G1/1/2: 10.0.0.70/30
- c. Loopback0: 10.0.0.81/32

9. Configure the following IP addresses on DSW-B2:

- a. G1/1/1: 10.0.0.58/30
- b. G1/1/2: 10.0.0.74/30
- c. Loopback0: 10.0.0.82/32

10. Manually configure SRV1's IP settings:

- a. Default Gateway: 10.5.0.1
- b. IPv4 Address: 10.5.0.4
- c. Subnet Mask: 255.255.255.0

11. Configure the following management IP addresses on the Access switches (interface VLAN 99), and configure the appropriate subnet's first usable address as the default gateway.

- a. ASW-A1: 10.0.0.4/28
- b. ASW-A2: 10.0.0.5/28
- c. ASW-A3: 10.0.0.6/28
- d. ASW-B1: 10.0.0.20/28
- e. ASW-B2: 10.0.0.21/28
- f. ASW-B3: 10.0.0.22/28

12. Configure HSRPv2 group 1 for Office A's Management subnet (VLAN 99). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

- a. Subnet: 10.0.0.0/28
- b. VIP: 10.0.0.1
- c. DSW-A1: 10.0.0.2
- d. DSW-A2: 10.0.0.3

13. Configure HSRPv2 group 2 for Office A's PCs subnet (VLAN 10). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

- a. Subnet: 10.1.0.0/24

b. VIP: 10.1.0.1

c. DSW-A1: 10.1.0.2

d. DSW-A2: 10.1.0.3

14. Configure HSRPv2 group 3 for Office A's Phones subnet (VLAN 20). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

a. Subnet: 10.2.0.0/24

b. VIP: 10.2.0.1

c. DSW-A1: 10.2.0.2

d. DSW-A2: 10.2.0.3

15. Configure HSRPv2 group 4 for Office A's Wi-Fi subnet (VLAN 40). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

a. Subnet: 10.6.0.0/24

b. VIP: 10.6.0.1

c. DSW-A1: 10.6.0.2

d. DSW-A2: 10.6.0.3

16. Configure HSRPv2 group 1 for Office B's Management subnet (VLAN 99). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

a. Subnet: 10.0.0.16/28

b. VIP: 10.0.0.17

c. DSW-B1: 10.0.0.18

d. DSW-B2: 10.0.0.19

17. Configure HSRPv2 group 2 for Office B's PCs subnet (VLAN 10). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

a. Subnet: 10.3.0.0/24

b. VIP: 10.3.0.1

c. DSW-B1: 10.3.0.2

d. DSW-B2: 10.3.0.3

18. Configure HSRPv2 group 3 for Office B's Phones subnet (VLAN 20). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

a. Subnet: 10.4.0.0/24

b. VIP: 10.4.0.1

c. DSW-B1: 10.4.0.2

d. DSW-B2: 10.4.0.3

19. Configure HSRPv2 group 4 for Office B's Servers subnet (VLAN 30). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

a. Subnet: 10.5.0.0/24

b. VIP: 10.5.0.1

c. DSW-B1: 10.5.0.2

d. DSW-B2: 10.5.0.3

Part 4 – Rapid Spanning Tree Protocol

- 1.** Configure Rapid PVST+ on all Access and Distribution switches.
 - a.** Ensure that the Root Bridge for each VLAN aligns with the HSRP Active router by configuring the lowest possible STP priority.
 - b.** Configure the HSRP Standby Router for each VLAN with an STP priority one increment above the lowest priority.
- 2.** Enable PortFast and BPDU Guard on all ports connected to end hosts (including WLC1). Perform the configurations in interface config mode.

Part 5 – Static and Dynamic Routing

- 1.** Configure OSPF on R1 (LAN-facing interfaces) and all Core and Distribution switches (all Layer-3 interfaces).
 - a.** Use process ID 1 and Area 0.
 - b.** Manually configure each device's RID to match the loopback interface IP.
 - c.** On switches, use the network command to match the exact IP address of each interface.
 - d.** On R1, enable OSPF in interface config mode.
 - e.** Make sure OSPF is enabled on all loopback interfaces, too. Loopback interfaces should be passive.
 - f.** Each Distribution switch's SVIs (except the Management VLAN SVI) should be passive, too.
 - g.** Configure all physical connections between OSPF neighbors to use a network type that doesn't elect a DR/BDR. NOTE: This doesn't work on the Layer-3 PortChannel interfaces between CSW1/CSW2. Leave them as the default network type.
- 2.** Configure one static default route for each of R1's Internet connections. They should be recursive routes.
 - a.** Make the route via G0/1/0 a floating static route by configuring an AD value 1 greater than the default.
 - b.** R1 should function as an OSPF ASBR, advertising its default route to other routers in the OSPF domain.

Part 6 – Network Services: DHCP, DNS, NTP, SNMP, Syslog, FTP, SSH, NAT

1. Configure the following DHCP pools on R1 to make it serve as the DHCP server for hosts in Offices A and B. Exclude the first ten usable host addresses of each pool; they must not be leased to DHCP clients.

a. Pool: A-Mgmt

- i. Subnet: 10.0.0.0/28
- ii. Default gateway: 10.0.0.1
- iii. Domain name: jeremysitlab.com
- iv. DNS server: 10.5.0.4 (SRV1)
- v. WLC: 10.0.0.7

b. Pool: A-PC

- i. Subnet: 10.1.0.0/24
- ii. Default gateway: 10.1.0.1
- iii. Domain name: jeremysitlab.com
- iv. DNS server: 10.5.0.4 (SRV1)

c. Pool: A-Phone

- i. Subnet: 10.2.0.0/24
- ii. Default gateway: 10.2.0.1
- iii. Domain name: jeremysitlab.com
- iv. DNS server: 10.5.0.4 (SRV1)

d. Pool: B-Mgmt

- i. Subnet: 10.0.0.16/28
- ii. Default gateway: 10.0.0.17
- iii. Domain name: jeremysitlab.com
- iv. DNS server: 10.5.0.4 (SRV1)
- v. WLC: 10.0.0.7

e. Pool: B-PC

- i. Subnet: 10.3.0.0/24
- ii. Default gateway: 10.3.0.1
- iii. Domain name: jeremysitlab.com
- iv. DNS server: 10.5.0.4 (SRV1)

f. Pool: B-Phone

- i. Subnet:** 10.4.0.0/24
- ii. Default gateway:** 10.4.0.1
- iii. Domain name:** jeremysitlab.com
- iv. DNS server:** 10.5.0.4 (SRV1)

g. Pool: Wi-Fi

- i. Subnet:** 10.6.0.0/24
- ii. Default gateway:** 10.6.0.1
- iii. Domain name:** jeremysitlab.com
- iv. DNS server:** 10.5.0.4 (SRV1)

2. Configure the Distribution switches to relay wired DHCP clients' broadcast messages to R1's Loopback0 IP address.

3. Configure the following DNS entries on SRV1:

- a.** google.com = 172.253.62.100
- b.** youtube.com = 152.250.31.93
- c.** jeremysitlab.com = 66.235.200.145
- d.** www.jeremysitlab.com = jeremysitlab.com

4. Configure all routers and switches to use domain name **jeremysitlab.com** and use SRV1 as their DNS server.

5. Configure NTP on R1:

- a.** Make R1 a stratum 5 NTP server.
- b.** R1 should learn the time from NTP server 216.239.35.0.
- c.** NOTE: NTP takes a LONG time to sync, especially in Packet Tracer. After making the configurations, you can move on – don't wait for the devices to sync.

6. All Core, Distribution, and Access switches should use R1's loopback interface as their NTP server.

- a.** Clients should authenticate R1 using key number **1** and the password **ccna**.

7. Configure the SNMP community string **SNMPSTRING** on all routers and switches. The string should allow GET messages, but not SET messages.

8. Configure Syslog on all routers and switches:

- a.** Send Syslog messages to SRV1. Messages of all severity levels should be logged.
- b.** Enable logging to the buffer. Reserve 8192 bytes of memory for the buffer.

9. Use FTP on R1 to download a new IOS version from SRV1:

- a. Configure R1's default FTP credentials: username **cisco**, password **cisco**.
- b. Use FTP to copy the file **c2900-universalk9-mz.SPA.155-3.M4a.bin** from SRV1 to R1's flash drive.
- c. Reboot R1 using the new IOS file, and then delete the old one from flash.

10. Configure SSH for secure remote access on all routers and switches.

- a. Use the largest modulus size for the RSA keys.
- b. Allow SSHv2 connections only.
- c. Create standard ACL 1, only allowing packets sourced from Office A's PCs subnet. Apply the ACL to all VTY lines to restrict SSH access.
- d. Allow only SSH connections to the VTY lines.
- e. Require users to log in with a local user account when connecting via SSH.
- f. Configure synchronous logging on the VTY lines.

11. Configure static NAT on R1 to enable hosts on the Internet to access SRV1 via the IP address **203.0.113.113.**

12. Configure pool-based dynamic PAT on R1 to enable hosts in the Office A PCs, Office A Phones, Office B PCs, Office B Phones, and Wi-Fi subnets to access the Internet.

- a. Use standard ACL 2 to define the appropriate inside local address ranges in the following order:
 - i. Office A PCs: 10.1.0.0/24
 - ii. Office A Phones: 10.2.0.0/24
 - iii. Office B PCs: 10.3.0.0/24
 - iv. Office B Phones: 10.4.0.0/24
 - v. Wi-Fi: 10.6.0.0/24
- b. Define a range of inside global addresses called **POOL1**, specifying the range 203.0.113.200 to 203.0.113.207 with a /29 netmask.
- c. Map ACL 2 to POOL1 and enable PAT. Confirm that hosts can access the Internet by pinging jeremysitlab.com.
- d. Verify that Internet link failover works by disabling R1's G0/0/0 interface and pinging again.
 - i. You will need to remove and re-configure the OSPF default-information originate command for this to work. In real Cisco routers, you can configure the default-information originate always command that supports failover like this, but the command isn't available in Packet Tracer.
 - ii. Re-enable G0/0/0 (and remove and re-configure default-information originate once again).

13. Disable CDP on all devices and enable LLDP instead.

- a.** Disable LLDP Tx on each Access switch's access port (F0/1).

Part 7 – Security: ACLs and Layer-2 Security Features

1. Configure extended ACL **OfficeA_to_OfficeB** where appropriate:

- a.** Allow ICMP messages from the **Office A PCs** subnet to the **Office B PCs** subnet.
- b.** Block all other traffic from the **Office A PCs** subnet to the **Office B PCs** subnet.
- c.** Allow all other traffic.
- d.** Apply the ACL according to general best practice for extended ACLs.

2. Configure Port Security on each Access switch's F0/1 port:

- a.** Allow the minimum necessary number of MAC addresses on each port.
 - i.** SRV1 does not use virtualization, so it uses a single MAC address.
- b.** Configure a violation mode that blocks invalid traffic without affecting valid traffic. The switches should send notifications when invalid traffic is detected.
- c.** Switches should automatically save the secure MAC addresses they learn to the running-config.

3. Configure DHCP Snooping on all Access switches.

- a.** Enable it for all active VLANs in each LAN.
- b.** Trust the appropriate ports.
- c.** Disable insertion of DHCP Option 82.
- d.** Set a DHCP rate limit of 15 pps on active untrusted ports.
- e.** Set a higher limit (100 pps) on ASW-A1's connection to WLC1.

4. Configure DAI on all Access switches.

- a.** Enable it for all active VLANs in each LAN.
- b.** Trust the appropriate ports.
- c.** Enable all optional validation checks.

Part 8 – IPv6

1. To prepare for a migration to IPv6, enable IPv6 routing and configure IPv6 addresses on R1, CSW1, and CSW2:

- a.** R1 G0/0/0: 2001:db8:a::2/64
- b.** R1 G0/1/0: 2001:db8:b::2/64

- c.** R1 G0/0 and CSW1 G1/0/1: Use prefix 2001:db8:a1::/64 and EUI-64 to generate an interface ID for each interface.
- d.** R1 G0/1 and CSW2 G1/0/1: Use prefix 2001:db8:a2::/64 and EUI-64 to generate an interface ID for each interface.
- e.** CSW1 Po1 and CSW2 Po1: Enable IPv6 without using the 'ipv6 address' command.

2. Configure two default static routes on R1:

- a.** A recursive route via next hop 2001:db8:a::1.
- b.** A fully-specified route via next hop 2001:db8:b::1. Make it a floating route by configuring the AD 1 higher than default.

Part 9 – Wireless

1. Access the GUI of WLC1 (<https://10.0.0.7>) from one of the PCs.

- a.** Username: admin
- b.** Password: adminPW12

2. Configure a dynamic interface for the Wi-Fi WLAN (10.6.0.0/24)

- a.** Name: Wi-Fi
- b.** VLAN: 40
- c.** Port number: 1
- d.** IP address: .4 of its subnet

Note: The Mega Lab video shows 10.6.0.2. This is a mistake, as it's a duplicate address of DSW-A1's VLAN 40 address.

- e.** Gateway: .1 of its subnet
- f.** DHCP server: 10.0.0.76

3. Configure and enable the following WLAN:

- a.** Profile name: Wi-Fi
- b.** SSID: Wi-Fi
- c.** ID: 1
- d.** Status: Enabled
- e.** Security: WPA2 Policy with AES encryption, PSK of cisco123

4. Verify that both LWAPs have associated with WLC1.

- a.** Due to Packet Tracer's limitations, wireless clients won't be able to lease an IP address from the Wi-Fi DHCP pool.